A Hands-on Lab on Agentic Scientific Discovery for Time Series with the Model Context Protocol

Dhaval Patel[1] **Miaomiao Feng**[2] **Chathurangi Shyalika**[3] **Shuxin Lin**[1] **Hasnat Md. Abdullah**[4]
[1]IBM T.J. Watson Research Center, Yorktown Heights, NY, USA
[2]Rensselaer Polytechnic Institute, Troy, NY, USA
[3]AI Institute, University of South Carolina, Columbia, USA
[4]University of Illinois Urbana-Champaign, Illinois, USA
pateldha@us.ibm.com, fengm4@rpi.edu, jayakodc@email.sc.edu, shuxin.lin@ibm.com, hma18@illinois.edu

Title. A Hands-on Lab on Agentic Scientific Discovery for Time Series with the Model Context Protocol.

Type. *Lab.* Every attendee installs a running system and drives it through five timed sprints. Laptops are required.

Sub-areas and keywords. Primary: Machine Learning (ML) Time-Series and Data Streams. Secondary: Multiagent Systems (MAS) LLM Agents and Tool Use; Search and Optimization (SO) Evolutionary Computation. Keywords: Model Context Protocol; time series foundation models; agentic scientific discovery; AutoML for time series; benchmark and evaluation design; industrial AI.

Presenters. Dhaval Patel (IBM, corresponding, pateldha@us.ibm.com); Miaomiao Feng (Rensselaer Polytechnic Institute, fengm4@rpi.edu); Chathurangi Shyalika (University of South Carolina, jayakodc@email.sc.edu); Shuxin Lin (IBM, shuxin.lin@ibm.com); Hasnat Md. Abdullah (UIUC, hma18@illinois.edu).

Suggested duration. Half day. We ask for the 3 hour 30 minute slot and use 180 minutes of instruction plus the 30-minute break, holding the remaining allotment as install-triage and overflow buffer.

Goal

An agent that calls `forecast(series)` is not doing science. A scientist looks at the data first, forms a hypothesis about what model class the signal deserves, composes an experiment, reads the diagnostics, and revises. This lab teaches attendees to build agents that work the second way, and to measure whether the reasoning was any good. The vehicle is a protocol question with a sharp answer: what does an MCP server have to expose so that a language agent can run a real model-discovery campaign over hundreds of time series estimators, rather than rubber-stamp a plan some AutoML system already made?

Attendees leave with three transferable design commitments and the evidence for each. *Components are data, not tools*: a catalogue of 716 estimators, detectors, transformers and extractors reaches the agent through 33 tools, because a model card is an *argument* to one generic `run_recipe` rather than a tool to dispatch on; long tool contexts degrade function selection badly (Kate et al. 2025). *Evidence, not decisions*: the server reports dominant period, stationarity, channel correlation and missingness, and deliberately reports no recommended lookback, so the parameter choice stays with the agent and stays gradeable. *Grade the chain, not the number*: a rubric the agent never sees scores its lookback, context, horizon and thresholding against references derived from the evidence. Reasoned choices score 1.00 against 0.40 for defaults on forecasting, and 0.67 against 0.00 on anomaly detection. Attendees reproduce both gaps within the first ninety minutes.

Copyright © 2027, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

Tools and Environment

The lab runs on TSM-MCP-Kit, an open server speaking the Model Context Protocol (https://modelcontextprotocol.io), shipped in the AssetOpsBench repository (Patel et al. 2026) under Apache-2.0. It is 13.4 kLOC over seven layers, 130 test functions, and 46 MCP tools of which 33 reach the agent and 13 administer the catalogue. sktime is the execution substrate, so every component resolves through one interface: 298 estimators, 150 transformers and 22 foundation-model adapters from sktime 1.1.0 (TTM, TSPulse, Chronos, MOIRAI, MOMENT, TimesFM, TimeMoE, PatchTST), 46 PyOD detectors carrying the AnomalyKiTS pipelines and thresholding (Patel et al. 2022), and 222 FLOps extractors. Eight standardized task contracts cover forecasting, regression, classification, anomaly detection, imputation, evaluation, similarity search and clustering.

Delivery is local Docker. Attendees run `docker compose up` on two published images, `assetopsbench-basic` and `assetopsbench-extra`, which carry the server, a CouchDB catalogue seed, the substrate and a curated slice of AssetOpsBench telemetry, so the room is offline-capable and conference wifi is off the critical path. Any MCP-speaking client attaches over stdio; we ship the ReActXen reference agent (Rayfield et al. 2025) plus adapters for common desktop clients. The classical and detector paths run CPU-only with no API key; we supply keys for the hosted foundation models. Images, compose file and a preflight script publish four weeks ahead, and the break doubles as an install clinic. Attendees keep the images, the catalogue and their run lineage afterwards.

Content Outline

Five modules, each ending in a timed sprint on the attendee's own machine. Module 1 installs the instrument and estab-

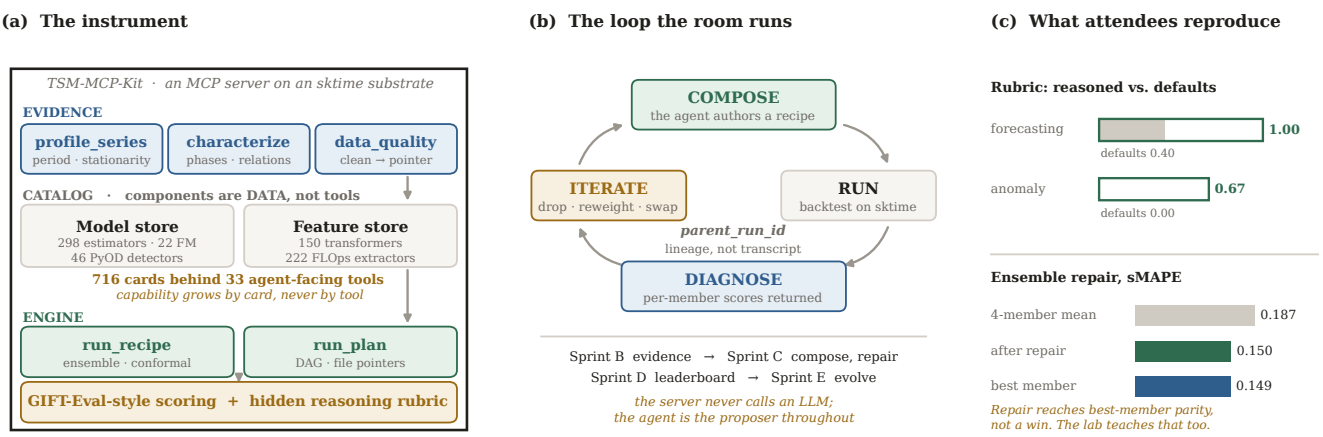


Figure 1: The lab in one view. (a) The instrument attendees install: evidence tools that state facts and decide nothing, 716 catalogue components exposed as data rather than tools, and a scoring layer that grades the decision chain. (b) The loop each sprint exercises, with the back edge that turns a refinement trajectory into persisted state. (c) The two result pairs every attendee reproduces on their own laptop, including the one that does not work.

lishes why a catalogue must be data rather than tools. Module 2 reads evidence and derives parameters from it, ending in a room-wide lookback vote the hidden rubric then grades. Module 3 introduces the Recipe and runs the compose–run–diagnose–iterate loop until the agent repairs its own ensemble from per-member diagnostics. Module 4 scores the campaign with a GIFT-Eval-style protocol (Aksu et al. 2024) and opens the rubric. Module 5 turns the loop into search with an AlphaEvolve-style ask/tell archive (Novikov et al. 2025). A closing module names what does not work yet. Figure 1 maps the parts.

Schedule

Time	Min	Module	Speaker
0:00	10	Act 0: the failing baseline	Patel
0:10	30	M1: The instrument *(Sprint A)*	Lin
0:40	30	M2: Evidence, not decisions *(Sprint B)*	Feng
1:10	30	*Break* and install clinic	all
1:40	40	M3: The Recipe and the loop *(Sprint C)*	Patel
2:20	30	M4: Scoring the scientist *(Sprint D)*	Shyalika
2:50	25	M5: Search as discovery *(Sprint E)*	Lin
3:15	15	Open problems, take-home challenge	Feng, all

History of Similar Offerings

The presenters have delivered six tutorials and labs on adjacent material: AAAI-23 and KDD-23 on anomaly detection at enterprise scale; AAAI-25, *Developing Explainable Multimodal AI Models*, 50+ attendees; AAMAS-25 on multiagent industrial copilots; AAAI-26, *From Inception to Productization: Hands-on Lab for the Lifecycle of Multimodal Agentic AI in Industry 4.0*, 100+ attendees; and KDD-26, *Building Reliable Industrial Agents with MCP*, accepted to the hands-on track.

This proposal succeeds the KDD-26 session rather than reprising it. KDD-26 teaches orchestration: a supervisory agent routing across IoT, work-order, failure-mode and document servers to finish an operations task. This lab narrows to one server and asks what KDD-26 never touches: how an agent runs a model-discovery campaign over a large component catalogue, and how anyone can tell whether its reasoning was sound. The catalogue-as-argument-space design, the evidence-versus-decisions split, the hidden rubric, the scoring protocol and the evolutionary archive are all new. We note for the chairs that a partly overlapping author set has submitted an independent tutorial proposal on evaluation methodology to this forum; the two are disjoint in content and we will gladly coordinate.

Estimated Number of Participants

We expect 60 to 90 attendees and cap hands-on registration at 100 so the sprints stay workable. Measured turnout anchors the estimate: 100+ at the AAAI-26 lab, 60+ at KDD-26, 300+ at the NeurIPS-25 expo demonstration of this codebase, and 149 teams in the AssetOpsBench challenge at CODS-COMAD 2025. Four presenters give one instructor per 20 attendees, the ratio that kept the AAAI-26 lab on schedule.

Prerequisite Knowledge

Attendees should be comfortable with Python and `docker compose`, and should have seen a language-model agent call a tool once, through MCP, LangGraph or a hand-rolled loop. Time series background helps but is not assumed: we define seasonality, lookback, context length, backtesting and MASE on first use, and the evidence tools report these quantities rather than asking attendees to compute them. No prior sktime, foundation-model or AutoML exposure is needed. A

laptop with 16 GB of memory, 20 GB of free disk and working Docker is required; no GPU, since the classical paths run on CPU and foundation models are served remotely.

Detailed Content

Act 0, the failing baseline (10 min). Before any teaching, the room runs one identical prompt against one chiller. The agent takes defaults, picks a context shorter than the seasonal cycle, and posts a score. Nobody is told why. The number goes on the board and stays there for three hours.

Module 1, the instrument (30 min). Why MCP, and what a server owes an agent. Three incumbent shapes, and what each costs. Tool-per-capability servers grow context with the library, and function selection degrades by 7 to 85 percent as those lists grow (Kate et al. 2025). AutoML-TS systems, our own AutoAI-TS (Shah et al. 2021) among them, decide the parameters for the user, so an agent built on them has nothing to reason about and nothing to grade. Model-routing agents pick among a handful of models, which the 2025 agentic time series survey records as the published state of the art (Chang et al. 2026). We teach the alternative: HuggingGPT's selection of models by description (Shen et al. 2023), pushed further so the catalogue becomes the argument space and bulk data crosses only as a file pointer. *Sprint A*: bring the server up, attach a client, call `list_tasks` and `discover_components`, and see 716 components arrive behind a tool list that does not grow.

Module 2, evidence, not decisions (30 min). The evidence tools and what they refuse to say. `profile_series` returns the dominant period computed on the detrended signal, seasonality strength, stationarity, inter-channel correlation and missingness. `characterize_series` returns per-phase state and rate across changepoints plus a bivariate relation label, and `data_quality` returns a cleaned file pointer. None returns a recommendation. We then walk the coupling that makes this a reasoning problem rather than a lookup: lookback follows the seasonal period, context length must cover the lookback, and the model follows the context, so the model cannot be chosen first. Context length behaves like a hyperparameter rather than a dataset property, which is exactly why the choice must be reasoned from evidence. *Sprint B*: each attendee derives a lookback and context for their asset; the room votes; we open the rubric's reference, one to three times the measured period, and each attendee scores their own run.

Module 3, the Recipe and the loop (40 min). The Recipe is the declarative object the agent authors: transforms with automatic inverse, a single estimator or an ensemble with a combiner, an optional conformal block, and an evaluation splitter. It compiles to one sktime forecaster, so an ensemble of a foundation model and a classical baseline is one line and the substrate executes it. `run_recipe` backtests and returns the aggregate score and, critically, the per-member scores. That is the diagnostic the agent iterates on. We run the loop live: a four-member mean ensemble scores sMAPE 0.187 while its members score 0.152, 0.662, 0.149 and 0.210, so the loop exposes the member dragging the result. The agent drops it, reweights, and calls `run_recipe` again with `parent_run_id` set, reaching 0.150. The repaired ensemble reaches parity with its best member at 0.149, not a win. Attendees watch a real loop produce a real non-result, and we discuss what would beat it. *Sprint C*: compose, read the per-member table, repair, and inspect the persisted lineage.

Module 4, scoring the scientist (30 min). How to score a campaign so one cherry-picked dataset cannot carry it. We adopt the GIFT-Eval protocol (Aksu et al. 2024): evaluate over many configurations of dataset, frequency, horizon and channel count, normalize each by a seasonal-naive baseline, aggregate by geometric mean, and rank by mean rank. We then open the hidden rubric and make the pedagogical point of the lab: the graded object is the decision chain, not the final number. We close on leakage, separating the temporal leakage that sequential splitting closes from the selection leakage it does not. *Sprint D*: teams submit recipes to a live leaderboard and defend a mean-rank win.

Module 5, search as discovery (25 min). The loop becomes a search when the archive remembers. We map AlphaEvolve (Novikov et al. 2025) onto the server: `evolve_ask` samples parent programs, diverse inspirations from other behaviour cells, and the evidence; the agent mutates or recombines them into one candidate; `evolve_tell` validates, evaluates to a scalar fitness, and places the candidate in a MAP-Elites archive keyed by task, regime, component count and ensemble shape; `evolve_best` returns the frontier. The server never calls a language model. Two things evolve: recipes, and feature programs with fit, transform and inverse, gated on invertibility before archiving. *Sprint E*: one mutation round each, merged into a shared archive on screen, so the room watches a frontier fill.

Open problems (15 min). We close on what the system cannot do, because a lab that only demonstrates successes teaches a bad habit. Exogenous and covariate forecasting is not on the recipe path, though Chronos-2 and MOIRAI-2 made covariates a headline feature. Evaluation is neither shift-aware nor streaming. Calibration coverage is computed but not scored. The ensemble reaches best-member parity rather than beating it. Multiagent orchestration across several such servers, the direction SmartPilot takes (Shyalika et al. 2025), sits outside this lab. Each gap becomes a take-home challenge against the public repository and the leaderboard, which stay open after the conference.

Lab Hands-on Component

The hands-on component is the lab, not an appendix to it: five sprints, 75 of the 180 instructional minutes, on the attendee's own machine. Sprints A and B establish the instrument and the evidence discipline; Sprint C runs the composition loop end to end, repair included; Sprint D is a competitive leaderboard round; Sprint E fills a shared evolutionary archive the room watches grow. Each sprint ships a green-path notebook so a failure in step one strands nobody, plus a stretch task and a checkpoint command that prints the attendee's rubric score and lineage, so instructors read the room at a glance.

References

Aksu, T.; Woo, G.; Liu, J.; Liu, X.; Liu, C.; Savarese, S.; Xiong, C.; and Sahoo, D. 2024. GIFT-Eval: A Benchmark for General Time Series Forecasting Model Evaluation. *arXiv:2410.10393*.

Kate, K.; Pedapati, T.; Basu, K.; Rizk, Y.; Chenthamarakshan, V.; Chaudhury, S.; Agarwal, M.; and Abdelaziz, I. 2025. LongFuncEval: Measuring the Effectiveness of Long Context Models for Function Calling. *arXiv:2505.10570*.

Novikov, A.; Vũ, N.; Eisenberger, M.; et al. 2025. AlphaEvolve: A Coding Agent for Scientific and Algorithmic Discovery. *arXiv:2506.13131*.

Patel, D.; Ganapavarapu, G.; Jayaraman, S.; Lin, S.; Bhamidipaty, A.; and Kalagnanam, J. 2022. AnomalyKiTS: Anomaly Detection Toolkit for Time Series. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(11): 13209–13211. Demonstration Program.

Patel, D.; Lin, S.; Rayfield, J.; Zhou, N.; Martinez, N.; d'Onnocha, R.; and Kalagnanam, J. 2026. AssetOpsBench: Benchmarking AI Agents for Task Automation in Industrial Asset Operations and Maintenance. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '26), Volume 2*, pages 9649–9660. https://doi.org/10.1145/3770855.381756.

Rayfield, J. T.; Lin, S.; Zhou, N.; and Patel, D. C. 2025. ReAct Meets Industrial IoT: Language Agents for Data Access. In *Proceedings of EMNLP 2025: Industry Track*, 364–382.

Shah, S. Y.; Patel, D.; Vu, L.; Dang, X.-H.; Chen, B.; Kirchner, P.; Samulowitz, H.; Wood, D.; Bramble, G.; Gifford, W. M.; Ganapavarapu, G.; Vaculin, R.; and Zerfos, P. 2021. AutoAI-TS: AutoAI for Time Series Forecasting. In *Proceedings of the 2021 ACM SIGMOD International Conference on Management of Data*.

Shen, Y.; Song, K.; Tan, X.; Li, D.; Lu, W.; and Zhuang, Y. 2023. HuggingGPT: Solving AI Tasks with ChatGPT and its Friends in Hugging Face. In *Advances in Neural Information Processing Systems 36*.

Shyalika, C.; Prasad, R.; El Ghazo, A.; Eswaramoorthi, D.; Muthuselvam, S. S.; and Sheth, A. 2025. SmartPilot: A Multiagent CoPilot for Adaptive and Intelligent Manufacturing. In *IEEE Conference on Artificial Intelligence (CAI)*, 1–8.

Chang, C.; Shi, Y.; Cao, D.; Yang, W.; Hwang, J.; Wang, H.; Pang, J.; Wang, W.; Liu, Y.; Peng, W.-C.; and Chen, T.-F. 2026. A Survey of Reasoning and Agentic Systems in Time Series with Large Language Models. *Transactions on Machine Learning Research*, June. https://openreview.net/forum?id=l3QW42g6u3.

Presenter CVs

Dhaval Patel is a Senior Technical Staff Member and Lead Research Scientist at IBM T.J. Watson Research Center, where he leads AI benchmarking and evaluation for industrial asset operations. He holds a Ph.D. in Computer Science from the National University of Singapore and was a tenure-track Assistant Professor at IIT Roorkee. He is the technical lead and architect of AssetOpsBench and DiagnosticIQ, and the principal author of TSM-MCP-Kit. His publication record spans NeurIPS, ICLR, ACL, EMNLP, AAAI, IAAI, KDD, SIGMOD and VLDB, with roughly 3,450 citations, an h-index of 18 and an i10-index of 36. He is a Senior Member of the IEEE (2019) and the ACM (2021), and mentors graduate students at Columbia University and RPI. Directly relevant prior work: AnomalyKiTS (AAAI-22 demonstration), AI Model Factory (AAAI-23 demonstration), AutoAI-TS (SIGMOD-21), AssetOpsBench-Live (AAAI-26 demonstration), and the AssetOpsBench agentic challenge at CODS-COMAD 2025 with 149 participating teams. Teaching record: tutorials and labs at AAAI-23, KDD-23, AAAI-25, AAMAS-25, AAAI-26 and KDD-26. In this lab he opens with Act 0 and teaches Module 3.

Chathurangi Shyalika is a Ph.D. candidate at the AI Institute, University of South Carolina, and an incoming Research Scientist at IBM, working on multimodal and agentic AI, time series analysis, and neurosymbolic AI for industrial applications. Her work bridges low-level sensor data and high-level reasoning for autonomous systems, including SmartPilot, a multiagent copilot for intelligent manufacturing (CAI 2025, AAMAS 2025), and rare-event prediction in manufacturing. She has co-organized tutorials and labs at AAAI-25, AAMAS-25, AAAI-26, US2TS and KDD-26, and brings substantial experience leading hands-on technical sessions for mixed academic and industrial audiences. In this lab she teaches Modules 2 and 4, the evidence discipline and the evaluation protocol.

Shuxin Lin is a Research Engineer at IBM Research with expertise in full-stack LLM application development, evaluation, and MCP implementation. She is a principal contributor to the AssetOpsBench codebase and to the MCP server surface this lab installs, and her work focuses on benchmarking language agents and integrating heterogeneous industrial data sources. She co-authored AssetOpsBench, the EMNLP-2025 industry-track work on chain-of-thought reasoning for industrial asset health monitoring, and PHMForge. As the engineer who built much of the server, she owns the two modules where attendees touch the software directly: Module 1, the install and the tool surface, and Module 5, the evolutionary archive. She also leads the install clinic during the break.

Miaomiao Feng is a Ph.D. candidate in Industrial and Systems Engineering at Rensselaer Polytechnic Institute, advised by Prof. Yinan Wang, working on engineering-driven machine learning for industrial systems. She holds a B.S. and an M.S. in Industrial Engineering from the South China University of Technology. She collaborates with the IBM team on AssetOpsBench and on diagnostic benchmarking for industrial assets, and her engineering-systems training is what the evidence module needs: the coupling between seasonality, lookback and model context is a signal-processing argument before it is an agent argument, and she teaches it as one. In this lab she teaches Module 2, the evidence discipline and the parameter reasoning that follows from it, and leads the closing module on open problems.

Hasnat Md. Abdullah is a Ph.D. student in Animal Sci-